

Updated Radiation Quality Factors and Biological Endpoints: A Synthesis (2014–2024)

Introduction

Radiation quality – the type and energy of ionizing radiation – has a profound influence on biological effectiveness. The relative biological effectiveness (RBE) is a key metric comparing different radiations' potency against a reference (typically γ -rays or X-rays). Over the past decade (2014–2024), numerous studies have updated RBE values for various radiation types and explored endpoints ranging from **clonogenic cell survival** to **normal tissue toxicity**, **tumor control**, and **carcinogenesis**. These efforts have refined “radiation quality factors,” which in radiological protection are encapsulated in radiation weighting factors (w_R) or quality factors (Q) for stochastic risk ¹. This reference document provides an in-depth, up-to-date synthesis of RBE data and mechanistic insights for **Protons, Carbon Ions, Alpha Particles, Neutrons, Mixed Radiation Fields, and Space Radiation (HZE ions)**. Each section reviews current RBE values (2014–2024), discusses relevant biological endpoints (e.g. clonogenic survival, normal tissue effects, tumor control probabilities, cancer induction), explains underlying mechanisms (e.g. LET dependence, DNA damage complexity, repair differences), and compares with historical values to illustrate the evolution of understanding. Comprehensive tables summarize RBE ranges for each radiation type and endpoint with source citations. A consolidated comparison table is provided at the end to contrast RBE ranges across all radiations.

(Note: “High-LET” refers to high linear energy transfer radiation, which generally yields higher RBE due to dense ionizations. Unless otherwise specified, RBE values are relative to high-energy photons (~MeV X-rays). Radiation weighting factors (w_R) for protection (stochastic effects) are distinguished from RBE for deterministic effects like cell killing ².)

Protons (Low-LET Radiation)

Updated RBE Values and Endpoints

Background: Therapeutic proton beams are considered low-LET radiation, historically assigned a generic RBE of 1.1 in clinical practice ³. This assumes a proton dose is ~10% more biologically effective than an equal photon dose. However, recent studies (2014–2024) show that proton RBE is *not* truly constant; instead, it varies with proton energy (depth), linear energy transfer, dose, and biological system ⁴. As protons slow down near the Bragg peak, their LET sharply rises, leading to increased RBE at the distal end of the beam

⁴.

Clonogenic cell survival: For the critical endpoint of cell reproductive survival (clonogenic assays), proton RBE ranges from approximately **1.1 in the entry region up to ~1.7 at the distal fall-off** of a spread-out Bragg peak (SOBP) ⁴. An extensive review by Paganetti (2014) found an average RBE ~1.15 at mid-SOBP for 2 Gy fractions, increasing from ~1.1 at the proximal entrance to ~1.35 near the SOBP distal edge and reaching ~1.6–1.7 in the distal dose fall-off when averaged across many cell lines ⁴. This gradient

underscores that the **assumption of a constant 1.1 RBE** is an oversimplification ⁵ ⁶ . Experimental studies confirm significant proton RBE elevation at the beam's end: for example, human cell experiments at 62 MeV showed pronounced cell-killing RBE increases in the distal region, with measured RBE values aligning with predictions from LET-based models ⁷ .

Normal tissue toxicity: For late-responding normal tissues (low α/β), proton RBE is also slightly elevated but generally remains close to unity in most practical scenarios. In vivo and *ex vivo* data suggest **proton RBE for normal tissue endpoints is on the order of 1.0–1.2**, meaning proton effects are only marginally stronger than photons on average ⁸ . Indeed, a broad analysis found that “endpoints other than clonogenic cell survival” (including various normal tissue and functional endpoints) were generally consistent with an RBE of ~ 1.1 ⁸ . Some normal-tissue studies indicate a modest RBE increase for late complications: e.g. proton RBE for spinal cord injury or other late toxicities may slightly exceed 1.0 in high-LET regions, but typically not beyond ~ 1.2 – 1.3 in clinically relevant dose ranges ⁸ . Overall, no dramatic normal-tissue RBE escalation has been observed for proton therapy – a reassuring fact for its therapeutic ratio.

Tumor control: Clinically, tumor control probability with proton therapy has been comparable to photons when matched for RBE-weighted dose, consistent with an average RBE near 1.1. Preclinical data likewise show **proton RBE for tumor cell kill ~ 1.1 – 1.2** in most systems ⁴ ⁸ . There is no evidence of significantly enhanced tumor control purely from proton *biology* (the main advantages of protons are physical dose localization). That said, specific conditions (e.g. very low dose per fraction, or tumors with low α/β ratio) could experience a slightly higher RBE. Experiments indicate RBE tends to increase as the x-ray α/β of the tissue decreases ⁹ . This suggests proton therapy might have a slight edge in treating very radioresistant tumors or hypofractionated regimens, though in practice the effect is small. Overall, current data do **not** show proton therapy achieving higher tumor cure rates than photons via biological means alone when properly accounting for dose ⁸ .

Carcinogenesis: The RBE for radiation-induced cancer from protons is not definitively known, but it is thought to be near unity. Epidemiological data are sparse (due to relatively recent use of proton therapy and low second-cancer statistics). For radiation protection, ICRP assigns protons a weighting factor of 2 (effectively assuming $\text{RBE} \approx 2$ for stochastic risk) ¹⁰ . This conservative factor accounts for some uncertainty and the production of secondary neutrons at high energies. However, radiobiological considerations suggest **proton RBE for cancer induction is close to 1** in most cases. High-energy protons are sparsely ionizing, and animal studies (e.g. space radiation research) often find high-energy protons to have similar carcinogenic effectiveness as γ -rays ¹¹ . Thus, while regulators use a factor ~ 2 , the actual cancer risk RBE of therapeutic proton beams at clinical energies is likely around 1–1.2. Research in the past decade continues to monitor second malignancies in proton patients to refine this understanding.

Mechanistic Basis and Historical Comparison

Protons are **low-LET charged particles** (initial LET ~ 1 – 3 keV/ μm in therapy beams) that gradually gain LET toward the end of range. The biological effectiveness of protons increases with LET because higher ionization density causes more complex DNA damage clusters that are harder to repair. Near the Bragg peak, proton tracks create denser ionizations similar to lower-energy electrons, resulting in more lethal damage per dose ⁷ . This underlies the rising RBE along the depth-dose profile ⁴ . The **dose dependence** of proton RBE is also notable: RBE tends to increase as dose per fraction decreases ⁹ . At lower doses, X-rays spare more cells via sublethal damage repair (shouldered survival curve), whereas protons (with

slightly higher LET) cause relatively more irreparable damage, effectively raising RBE. Similarly, cells or tissues with greater capacity for repair (e.g. late-responding tissues with low α/β) exhibit a larger RBE, since photon damage can be significantly repaired between fractions, but proton damage (being somewhat more clustered) is less mitigated ⁹. These trends were already recognized historically, but recent modeling and experiments (2014–2020) have quantified them better. For instance, model-based studies and clonogenic assays show **proton RBE increases when treating radioresistant cells or using hypofractionation**, reinforcing the classical radiobiological principle that high-LET radiations negate some repair advantage ⁹.

Historically, proton quality factors in protection were set much higher – e.g., the 1977 ICRP recommendations gave protons a quality factor $Q = 10$ and ICRP 1990 reduced it to 5 ¹². By ICRP Publication 103 (2007), this was lowered to a weighting factor $w_R = 2$ for protons ¹⁰, reflecting consensus that protons are only mildly more effective than photons. Clinically, the use of $RBE = 1.1$ for proton therapy dates back to the 1970s and was affirmed by experimental averages ⁴. The **current understanding (2014–2024)** refines this by acknowledging **RBE variability** rather than a single value: the broad average is ~ 1.1 – 1.15 , but with a spread from ~ 0.9 – 1.7 depending on conditions ⁴. Modern treatment planning research is exploring incorporating variable RBE maps. For example, variable-RBE proton dose calculations have been proposed to account for the distal-end RBE rise and possibly “LET painting” to exploit RBE increases in hypoxic tumor subregions ¹³. These developments highlight how our theoretical understanding (LET and track structure effects) has translated into potential clinical practice changes. The field continues to emphasize that while **proton RBE variations are real, large uncertainties remain**, and a cautious approach (still assuming ~ 1.1 for safety with margin) is used pending further in vivo validation ⁸.

Proton RBE Summary Table

Below is a summary of reported RBE ranges for protons across key endpoints, with experimental context and sources:

Biological Endpoint	Reported Proton RBE (range)	System / Conditions	Sources
Clonogenic cell survival	1.1–1.7 (typical: ~ 1.1 entrance; ~ 1.6 – 1.7 at distal Bragg peak)	In vitro mammalian cells along SOBP (2 Gy/fraction)	Paganetti 2014 review ⁴ ; Chaudhary <i>et al.</i> 2014 ⁷
Normal tissue toxicity (late effects)	~ 1.0–1.3 (usually ≈ 1.1)	Various in vivo endpoints (CNS, spinal cord, etc.) at therapeutic doses	Paganetti 2014 (aggregate of endpoints) ⁸ ; see also Marshall <i>et al.</i> 2016 (noted no large deviations) ¹³
Tumor control / regression	~ 1.0–1.2 (no clinically significant difference vs photons)	Clinical tumor outcomes (when physical dose adjusted by 1.1) and preclinical tumor models	Paganetti 2014 ⁸ ; see Britten <i>et al.</i> 2013 (in vitro tumor cell kill) ¹⁴

Biological Endpoint	Reported Proton RBE (range)	System / Conditions	Sources
Carcinogenesis (cancer induction)	~1.0 (assumed); <i>Protection</i> $w_{R} = 2$	Long-term second cancer risk (limited human data); space proton exposures in animals	ICRP 103 recommendation ¹⁰ ; ND internal NASA studies (proton risk $\approx$ photon) ¹¹

Carbon Ions (Intermediate-LET Radiation)

Updated RBE Values and Endpoints

Background: Carbon ions are high-LET radiation within the energy range used for radiotherapy (typically 150–400 MeV/nucleon). Carbon-ion radiotherapy (CIRT) deliberately exploits an **elevated RBE** in the Bragg peak to potentially widen the therapeutic window ¹⁵ ¹⁶ . Unlike proton therapy (with a fixed RBE=1.1), carbon therapy planning uses variable RBE models (e.g. Local Effect Model or Microdosimetric Kinetic Model) to calculate RBE-weighted dose ¹⁷ . Over 2014–2024, substantial data from centers in Japan and Europe have refined carbon RBE values for different tissues, doses, and fractionation schemes.

Clonogenic cell survival: Carbon ions produce far more biological damage per dose in the Bragg peak region than photons or protons. Reported **RBE for cell killing by carbon ions ranges roughly from ~2 up to 5 or more**, depending on LET and biological system ¹⁸ ¹⁹ . In typical clinical scenarios (spread-out Bragg peak covering a tumor), the RBE in the **target region** is on the order of 2–3 ²⁰ . Experimental cell survival curves confirm that carbon RBE increases with depth (LET): for instance, Chinese hamster cells irradiated in a water phantom showed similar survival in entrance for protons vs carbon, but markedly lower survival in the carbon target region, consistent with an RBE advantage ~2–3 ²¹ ²² . Under extreme conditions (e.g. very low doses or specific cell cycle phases), carbon-ion RBE can be higher. A recent radiobiology review noted that across many systems, carbon RBE can span **1 to 10**, with the upper end observed in special cases like very low doses or particularly radiosensitive endpoints ¹⁸ ¹⁹ . However, for practical therapy doses, carbon's RBE does **not** approach those extreme values – it stays moderate. Notably, **carbon's RBE peaks in the mid-LET range (~100 keV/μm)**, beyond which ultra-high LET causes “overkill” (wasted energy with no extra biological effect). The highest RBE for carbon is thus achieved at the distal edge of the SOBP (where residual carbon ion energy is lowest but still above fragmentation threshold). In summary, carbon therapy provides roughly **2–3 times greater cell-killing effectiveness** at the tumor Bragg peak compared to conventional X-rays ²⁰ , which is a principal rationale for its use.

Normal tissue toxicity: Carbon ions, being high-LET, also pose a risk of increased normal tissue damage, especially for late-responding tissues. Both animal experiments and patient data have been examined to quantify this. The **RBE for late effects** (e.g. spinal cord myelopathy, fibrosis) with carbon ions can be **1.5 to 3+**, depending on LET and dose per fraction ²³ ²⁴ . A 2019 study of rat spinal cord irradiation with carbon SOBP found RBE (at 50% myelopathy incidence) rising from ~1.46 at the shallow, low-LET position to ~3.26 at the deep high-LET position (for 6 × fractionated doses) ²³ . This illustrates that carbon's RBE for normal tissue increases with LET similarly to its tumor RBE. Importantly, that same study noted the RBE variation with fractionation was modest over the tested range – meaning the LET effect dominates over fractionation effect for carbon RBE ²⁴ ²⁵ . Clinically, carbon therapy centers constrain doses such that **normal tissue complication probabilities remain acceptable, using RBE-weighted dose calculations**. The entrance

region (higher dose to normal tissue but low LET) has an RBE closer to 1, while at depth inside or beyond the tumor, where stray dose is low but LET is high, one must consider risk of late effects (e.g. in distal normal tissues). Fortunately, for scanning carbon beams, secondary neutron production (which could deliver high-LET dose outside the field) is very low ²⁶. Therefore, the main normal tissue concern is immediately adjacent to the target where carbon LET is moderate. Available clinical data indicate that **carbon therapy's late toxicities are manageable and generally in line with modeled RBE**, provided careful planning. For example, the rate of serious late effects at NIRS and European centers has not exceeded what the RBE models predicted, suggesting current RBE models (LEM, MKM) are reasonably accurate for normal tissue endpoints ¹⁷.

Tumor control: Carbon ions were introduced to improve control of tumors that respond poorly to photons (e.g. hypoxic, slow-cycling, radioresistant histologies). The expectation is that an RBE of 2–3 in the tumor will allow a higher effective tumor dose without increasing normal tissue dose equivalently ¹⁵ ²⁷. Indeed, **clinical outcome data (2010s)** have shown promising control rates in certain diseases (e.g. skull-base chordomas, sarcomas) with carbon therapy, in some cases superior to historical photon results. While many factors influence these outcomes, radiobiological effectiveness is believed to play a role. Preclinical studies support that carbon ions have a higher tumor cell kill per Gy, especially for cells in hypoxic conditions or with low intrinsic radiosensitivity. For instance, in vitro studies on quiescent hypoxic tumor cells found carbon RBE up to ~4–5, significantly higher than in oxygenated conditions ²⁸. Animal tumor models (e.g. irradiated rat tumors) have shown carbon RBE in the range ~2–3 for endpoints like tumor growth delay or TCD₅₀ (dose for 50% control) ¹⁸. Therefore, **the RBE for tumor control with carbon ions is roughly 2–3** under clinical conditions, aligning with cell survival data. This means a carbon dose of e.g. 3 Gy (physical) can achieve similar tumor cell kill as ~6–7 Gy of photons in the high-LET region. It's important to note that carbon RBE is not uniform across the tumor volume – it is lower in the entry (plateau) region and highest at the distal end. Treatment planning algorithms ensure the *RBE-weighted dose* is homogeneous, which requires delivering a non-uniform physical dose (slightly higher physical dose in entrance, lower in distal layers) ²⁹. Clinically, this results in a uniform biologically effective dose that covers the tumor. Comparisons of carbon vs photon plans suggest only **small differences in predicted second cancer risk** and normal tissue sparing when proper RBE models and dose distributions are applied ³⁰ ³¹. Thus, improved tumor control with carbon is achieved mainly by the ability to escalate dose to the tumor (due to favorable dose distribution and RBE) rather than an *across-the-board* drastic increase in RBE.

Carcinogenesis: High-LET radiation like carbon can induce cancers more effectively per dose than photons, especially at low doses. Experiments have demonstrated that carbon ions have higher mutagenic and carcinogenic potential in cell and animal models ³². For example, carbon-ion exposure in mice yields increased chromosomal aberrations and tumor incidence relative to X-rays for the same absorbed dose ³². The **RBE for cancer induction by carbon** depends strongly on dose: at low doses (where photon exposure might have little effect), carbon's efficiency is much higher, yielding RBE values that can be very large (perhaps on the order of 5–10 or more) ¹⁹ ³³. However, at the moderate to high doses used in therapy, the difference in carcinogenic effectiveness between carbon and photons diminishes. As dose increases, the incremental RBE for cancer induction drops, partly because high doses of photons can also induce significant damage (the dose-response curves start converging) ³⁴. Importantly, in a clinical context, patients treated with carbon ions have not shown a drastic increase in secondary cancers compared to photon or proton patients, albeit follow-up times are limited. A recent review concluded that *“although C-ions are more effective than X-rays in cancer induction, no significant differences are observed in clinical settings”* ³⁵ ³⁶. This implies that any elevated risk might be subtle or offset by the fact that carbon treatments generally involve fewer beams and lower integral dose to normal tissue. Additionally, modern

active scanning delivers very low neutron stray dose ²⁶, reducing exposure of uninvolved tissue. Current radiation protection standards assign **heavy ions (including carbon) a quality factor of 20** for stochastic risk – same as α -particles ³⁷. Experimentally, carbon's carcinogenic RBE is somewhat lower than that of heavier ions or neutrons ³². For instance, neutron or iron-ion exposures produce higher tumor rates per Gy than carbon in many models ³². In summary, **carbon ions have a higher RBE for carcinogenesis than photons, but lower than the most densely ionizing radiations**. Ongoing animal studies are further examining carbon vs photon second-cancer risk, but current data do not warrant special restrictions beyond those accounted for by existing weighting factors.

Mechanistic Basis and Historical Comparison

Carbon ions exemplify **intermediate-to-high LET radiation** in therapy. At therapeutic energies, carbon LET in the plateau is ~10–30 keV/ μ m, rising to ~80–100+ keV/ μ m at Bragg peak. This LET range produces dense ionization tracks causing complex DNA double-strand breaks (DSB) in clusters. Mechanistically, carbon ions induce more difficult-to-repair damage (e.g. clustered DSBs with local complex lesions) than sparsely-ionizing radiation ³⁸ ³⁹. The **track structure** of carbon (continuous densely ionizing path) results in a higher probability of lethal lesions per particle traversal. Classic radiobiology from Barendsen and others showed that RBE reaches a maximum at LET ~100 keV/ μ m – interestingly, carbon Bragg-peak LET falls in this vicinity, which is a reason carbon was chosen for therapy ⁴⁰. Carbon's **theoretical advantages** include: (1) **Increased direct cell kill** – one carbon ion traversal through a nucleus can inactivate a cell, whereas many photons are needed (carbon has a small or absent “shoulder” on the cell survival curve) ⁴¹ ⁴²; (2) **Reduced oxygen effect** – carbon ions have a lower oxygen enhancement ratio (OER ~1.5 or less at high LET, vs OER ~2.5–3 for X-rays), meaning they kill hypoxic cells more efficiently; (3) **Less variation with cell cycle** – high-LET damage is effective even in radioresistant cell cycle phases. These factors collectively contribute to carbon's elevated RBE and potential to control resistant tumors. As **Durante et al. (2020)** note, carbon's LET in therapy is “moderate” compared to α particles or heavy Fe ions – enough to boost effectiveness but not so high as to cause purely untreatable normal tissue damage ⁴³ ⁴⁴. In fact, in the tumor volume of a typical carbon treatment, RBE is capped around ~3 and OER > 1 (some oxygen effect persists) ⁴³. If one wanted to exploit even higher-LET effects (e.g. RBE > 5 or OER ~1), heavier ions like oxygen or silicon would be needed ⁴⁵. There is ongoing research on using such ions or LET “painting” to further enhance tumor kill, but these approaches may incur higher normal tissue toxicity ⁴⁶.

Historically, carbon-ion radiobiology builds on extensive experiments at Lawrence Berkeley Lab (LBL) in the 1970s–80s and later at HIMAC (Japan) and GSI (Germany) ¹⁷. Early cell experiments established carbon RBE roughly 2–5 depending on cell type and endpoint. Clinicians at NIRS empirically adopted an RBE-weighted dose unit “Gy(Eq)” (later Gy(RBE)) to prescribe carbon doses; their clinical RBE model (MKM) assumed a reference RBE of ~3 at the Bragg peak for certain cell lines. Over time, models were refined, and in 2010s the ICRU recommended using just “Gy” for RBE-weighted dose (with clarity that it's RBE-adjusted) ⁴⁷ ⁴⁸. The **2014–2024 period** has seen confirmation of earlier RBE findings with more sophisticated tools (e.g. high-throughput omics, advanced animal models). It also became clear that carbon's **RBE changes along the SOBP must be accounted for**: a uniform biological dose requires a non-uniform physical dose distribution ⁴⁹, which all modern treatment planning systems handle. In comparison to **neutrons** (the first high-LET used clinically), carbon ions are more localized and controllable; yet the neutron experience foreshadowed carbon's behavior. In fact, neutron therapy taught that high-LET beams have **higher RBE for late tissues than early effects** and that RBE increases as dose per fraction decreases ⁵⁰ ⁵¹. Carbon ions adhere to these same principles (though to a lesser degree than fission neutrons). The result is that while carbon ions improve the *therapeutic ratio* by increasing tumor RBE in the target, one must carefully manage

the dose to surrounding tissues, which might experience a higher RBE for late injury than initially assumed if one naively used a constant factor ⁵⁰. Fortunately, decades of research have yielded sophisticated RBE models (LEM, MKM, etc.) that are continuously validated. Recent comparisons (LEM I vs LEM IV, etc.) against in vivo data (e.g. rat spinal cord) show these models can predict observed RBE within a reasonable accuracy ⁵² ⁵⁵. Still, uncertainties exist (e.g. variation between individuals, molecular pathways activated by high LET). The current trend is toward **biologically optimized treatment planning** – for example, considering not just RBE but also molecular endpoints and patient-specific factors (genomic markers for radiosensitivity) ⁵³ ⁵⁴. Overall, the past decade has solidified carbon ions' place as a potent modality with a quantifiable RBE advantage, while also reinforcing that this advantage must be wielded with knowledge and respect for underlying radiobiology.

Carbon Ion RBE Summary Table

Key RBE ranges for carbon ions by endpoint are summarized below, highlighting typical values and conditions:

Biological Endpoint	Reported Carbon RBE (range)	System / Conditions	Sources
Clonogenic cell survival	~2–3 (in SOBP target for 2 Gy/ fx); up to ~4–5 in some conditions	In vitro cell assays (oxygenated); higher in hypoxic or low-dose scenarios	Durante <i>et al.</i> 2020 (RBE ~1–3 in tumor volume) ⁵⁵ ; Incerti <i>et al.</i> 2020 (range 1–10 observed, context-dependent) ¹⁸ ¹⁹
Normal tissue toxicity (late effects)	~1.5–3.5 (increasing with LET)	E.g. rat spinal cord myelopathy, fractionated carbon SOBP (LET 16→99 keV/μm)	Karger <i>et al.</i> 2019: RBE 1.46 at 16 keV/μm to 3.26 at 99 keV/μm ²³ ²⁴ ; Durante 2020 (normal tissue RBE moderate, manageable) ²⁶
Tumor control (in vivo)	~2–3 (typical in treatment)	Clinical tumor response and animal tumor models (e.g. TCD₅₀)	Durante 2020 (carbon is “moderate LET,” RBE <3 in tumor) ⁵⁵ ; NIRS clinical reports (improved control in radioresistant tumors consistent with RBE~2–3, refs. therein)
Carcinogenesis (cancer induction)	~1–10 (dose-dependent; ~5–10 at low dose, ~1–2 at high dose)	Animal studies of tumor induction at low vs high doses; epidemiological projection	Durante 2020 (range ~1–10 depending on dose, lower than neutrons/Fe) ¹⁹ ³⁴ ; Incerti 2020 (carbon more effective than X for cancer, but clinical impact minimal) ³⁵ ³⁶

Alpha Particles (High-LET Radiation)

Updated RBE Values and Endpoints

Background: Alpha particles (such as those emitted by radionuclides like ^{238}Pu , ^{210}Po , ^{226}Ra , etc.) are **very high-LET** ($\approx 50\text{--}200\text{ keV}/\mu\text{m}$) radiation that deposit densely ionizing tracks over short ranges (tens of micrometers). Alphas historically have been assigned an extreme quality factor ($Q = 20$) in radiation protection due to their potent biological damage ³⁷ ⁵⁶. In recent years, interest in α -particles has been renewed via *targeted alpha therapies* (TAT) in oncology, where radionuclides emitting α 's are delivered to tumors. From 2014–2024, research on α -particle radiobiology has expanded, though much foundational knowledge remains consistent with earlier findings. Key features of α radiation include: a **high RBE for cell killing**, minimal dose-rate or fractionation sparing, and a high efficiency in inducing stochastic effects like mutations and cancers at low doses.

Clonogenic cell survival: Alpha particles are extraordinarily effective at killing cells. It typically takes only one or a few alpha tracks through a cell nucleus to inactivate the cell (contrast with $\sim$ multiple photon traversals needed). Experimentally, **RBE values for cell killing by α -particles are on the order of 3–7** (with variation by cell line and reference radiation) ⁵⁷. Seminal studies by Barendsen in the 1960s showed RBE peaks for LET $\sim 100\text{ keV}/\mu\text{m}$, aligning with typical alphas ⁵⁸ ⁵⁹. More recent evaluations continue to find high RBE: for instance, a Department of Energy panel reviewing α -particle literature recommended an RBE between **3 and 5** for cell killing (relative to cobalt-60 γ -rays) ⁵⁷. They specifically suggested using $\text{RBE} \approx 5$ as a reasonable planning value for deterministic effects of α -emitters ⁵⁷ ⁶⁰. Many in vitro studies with cultured mammalian cells report alpha RBE in this range for survival endpoints. Some specific examples: human fibroblasts exposed to ^{210}Po α particles had an RBE ~ 5 (for 10% survival level) compared to X-rays ⁶¹ ⁶²; V79 hamster cells exposed to 5 MeV alphas vs 250 kVp X-rays yielded RBE $\sim 6\text{--}7$ at high survival levels (since X-ray survival has a wide shoulder, the initial slope RBE of alphas can be even higher). In summary, **alpha particles generally have an RBE roughly 5 (± 2)** for cell inactivation, depending on conditions. This means an absorbed dose of 1 Gy of alphas can produce similar cell kill as ~ 5 Gy of conventional radiation, a dramatic difference.

Normal tissue toxicity: There is limited direct human data on normal tissue effects of external α -particles because they are not used as external beams clinically (due to their short range). However, insights come from accidental internal exposures, animal experiments, and projection from cell data. Because of the lack of fractionation sparing (discussed below), any normal tissue hit by α -particles will respond as if it received a single high-LET dose. The DOE panel (1996) recommended using **$\text{RBE} \approx 5$** for deterministic (normal tissue) effects in scenarios of internal α exposure ⁵⁷. This was a conservative approach in the absence of human data. Animal studies with TAT agents (e.g. ^{213}Bi , ^{211}At) have observed significant normal organ toxicity (to bone marrow, kidney, etc.) consistent with high RBE. For instance, myelotoxicity is often dose-limiting in α therapies at levels that correspond to a much higher biologic effect than the same dose of photons would cause. Specific data: relative to external beam, the bone marrow toxicity of α -emitters suggests an RBE on the order of 4–7 (though complicated by non-uniform dose distributions) ⁵⁶ ⁶³. In general, one can assume **alpha RBE for late tissue injury is similar ($\sim 3\text{--}5$)** as for cell kill, unless tissue-specific factors modify it. Notably, in radiobiology experiments, **fractionation does not spare normal tissues with α -particles** – splitting an alpha dose into parts yields nearly the same damage, indicating each particle track causes irreparable harm ⁶⁴ ⁶⁵. Thus, conventional normal-tissue sparing tactics (fractionation, protraction) are ineffective for alphas, reinforcing their high RBE.

Tumor control: The extremely high linear energy transfer of α -particles makes them attractive for killing malignant cells, including those resistant to low-LET radiation. In targeted alpha therapy (e.g. ^{223}Ra for bone metastases, ^{211}At -labeled antibodies for micrometastases), remarkable tumor responses have been observed, often eradicating tumor cells with minimal opportunity for repair between decays. While it is hard to assign a single RBE to “tumor control” by alphas (since these therapies involve non-uniform dose deposition at the cellular level), we can infer that **the RBE for tumor cell kill is similarly high (approximately 3–5 or more)**. For example, in a comparative experiment using an α -emitter vs γ -rays on a leukemia model, the dose required for tumor control with alphas was several-fold lower in Gy than for X-rays, consistent with RBE $\sim 5\text{--}7$ (though heterogeneity of dose complicates precise numbers) ⁶⁶ ⁶⁷. Clinically, the **efficacy of alpha therapies in otherwise radioresistant cancers** (like castrate-resistant bone metastases responding to ^{223}Ra) serves as qualitative evidence of their high RBE. In summary, one can surmise that to achieve the same tumor control probability, an alpha absorbed dose might be on the order of 1/5th (or less) of the photon dose. This is a double-edged sword: it means a small number of alpha decays in a tumor cell can sterilize it, but stray decays in normal tissue can likewise ablate those cells. Thus, the therapeutic index depends on targeted delivery. In external beam context (hypothetical), a RBE for tumor control would likely mirror the cell survival RBE (~ 5).

Carcinogenesis: Paradoxically, while extremely lethal at high doses, alpha radiation is also very potent at inducing cancers at low doses (in surviving cells). Epidemiological and animal data historically established that **α -particles have a high RBE for carcinogenesis**, which is why the radiation weighting factor $w_R = 20$ was chosen ⁵⁶ ⁶³. For example, studies of miners exposed to radon (an α -emitter) showed substantially higher lung cancer rates per unit dose than expected from γ -ray data. The prevailing estimate was that “per unit absorbed dose, α -particles are associated with a 20-fold greater risk of cancer induction than photons” ⁵⁶ ⁶³. However, recent re-evaluations of specific cancer types suggest the picture is more nuanced. A 2010 review of human and animal data questioned whether $Q = 20$ is universally appropriate ⁶⁸. For bone cancer induction by incorporated α -emitters, the RBE might be lower: when one considers microdosimetric bone dose distributions, the RBE for bone cancer risk was estimated in the range **3–12** instead of 20 ⁶⁹ ⁷⁰. Likewise, for leukemia induction (from bone marrow exposure), analyses indicated an RBE closer to **2–3** for alphas ⁷¹ ⁷². These findings imply that the standard factor of 20 may overestimate some risks or at least is not a one-size-fits-all. Nevertheless, the consensus remains that α -particles are far more carcinogenic per Gy than low-LET radiation, especially for solid epithelial cancers (lung, liver, etc.). At low doses, the RBE for cancer can be extremely high because a single alpha traversal can cause complex genetic damage leading to malignancy, whereas photons need multiple hits. Interestingly, the **“inverse dose-rate effect”** is pronounced: lowering the dose rate of alphas does *not* reduce cancer risk the way it does for photons, because even spaced-out alpha hits cause irreparable initiating damage ⁷³ ⁷⁴. In radiation protection, the use of quality factor 20 for alphas is still considered prudent, but ongoing research (2014–2024) is fine-tuning risk models. For instance, new biophysical models of marrow dosimetry and DNA repair are being used to update leukemia risk from inhaled alpha emitters, potentially leading to adjustments in effective quality factors. Overall, **alpha particles remain some of the most carcinogenic forms of radiation known**, and their high RBE for stochastic effects is well documented, even if the exact numerical value varies by endpoint and situation.

Mechanistic Basis and Historical Comparison

Alpha particles are the paradigm of **densely ionizing radiation**. An alpha (typically 4–8 MeV from radionuclides) deposits a large amount of energy in a very short path: its LET starts very high (even exceeding 100 keV/ μm near the end of its short range) ⁵⁸ ⁵⁹. Mechanistically, an alpha track produces a

core of ionizations and excitations that can shatter the DNA within its path. **DNA damage by alphas** often manifests as multiple double-strand breaks in proximity, complex chromosome rearrangements, and clustered lesions that overwhelm the cell's repair machinery ⁷⁵ ⁷⁶. Classic experiments (e.g. by Zirkle in 1930s, then Barendsen in 1960s) demonstrated that if an alpha particle's Bragg peak traverses a cell nucleus, the cell's ability to divide is destroyed at much lower doses than with X-rays ⁵⁸ ⁵⁹. The **linear-quadratic model** essentially breaks down for alpha radiation: cell survival curves are nearly exponential (linear) with little to no shoulder ⁷⁷ ⁷⁸. One high-LET hit can be a "single lethal event," whereas low-LET needs the accumulation of sublethal events ⁴¹ ⁴². This is why fractionation fails to spare cells – whether you give a dose in one go or in parts, the alpha-induced lethal events do not "heal" in between ⁶⁴ ⁷⁹. This was **experimentally confirmed** by splitting alpha doses: e.g. human kidney cells showed the same survival whether a dose was given all at once or in two fractions separated by hours ⁶⁴ ⁸⁰. The rapid repair of sublethal damage that photons enjoy simply doesn't apply for alphas (because they induce little sublethal, mostly lethal damage). A 2023 study combining X-rays and α -particles found that α radiation generates much more "sublethal damage" than X-rays (per dose) – interestingly, this led to **synergistic effects** when cells were exposed to mixed X + α (the alpha damage served as a load of sublethal lesions that X-ray could then convert to lethal) ⁸¹ ⁸². This underscores the unique damage signature of alphas.

Historically, the high RBE of alphas has long been recognized. The radium dial painters tragedy in the early 20th century and studies on Thorotrast patients (Thorium dioxide contrast, an α -emitter) demonstrated severe health effects (bone malignancies, liver cancers) from alpha doses that, in gray, were not extremely high. By mid-20th century, radiobiologists had quantified alpha RBE in cell systems and found values around 5–7 for cell killing, which influenced radiation protection guidelines. In the 1970s, ICRP set the quality factor for alphas at 10, later revising to 20 as more data on their carcinogenicity accumulated ¹². The 2014–2024 period hasn't so much changed the fundamental understanding of alpha RBE as **extended it to modern contexts**. For example, targeted alpha therapy experiments must effectively calibrate dosing and toxicity – recent MIRD publications compiled alpha RBE data to guide therapy, reiterating ~5 as a reference for deterministic effects ⁵⁷. There has also been progress in **microdosimetry** for alphas: since energy deposition is stochastic at cellular scale, RBE can vary with microdistribution. Work by Humm, Howell, and others in the 2000s–2010s explored how non-uniform distributions of α -emitters in tissue can change effective RBE (some cells get multiple hits, others none) ⁸³ ⁸⁴. But generally, those complexities don't change the conclusion that any cell directly traversed by an alpha is likely doomed. On the stochastic side, advancements in genomics and understanding of bystander effects also touched alpha radiobiology: interestingly, high-LET alphas can cause "bystander" damage in neighboring cells through chemical signals, but photons do as well – RBE for these secondary phenomena is still being studied. In radiation protection, questions have been raised (as mentioned) about whether Q = 20 is appropriate for all cancer types. The latest ICRP reports maintain the value 20 for simplicity, but acknowledge exceptions (like bone cancer from incorporated α , where microdosimetric averaging over bone might dilute the dose to target cells, yielding lower apparent RBE) ⁶⁹ ⁷⁰. Looking forward, **alpha particles remain a double-edged sword**: biologically superb at killing cancer cells, but with zero margin for error if normal cells are hit. Modern TAT trials are attempting to harness this with precision targeting at the molecular level.

Alpha Particle RBE Summary Table

The table below summarizes RBE ranges for α -particles across endpoints, drawing on both historical benchmarks and recent analyses:

Biological Endpoint	Reported α -Particle RBE (range)	System / Conditions	Sources
Clonogenic cell survival	~3–7 (reference ~5)	In vitro cell culture (various cell lines, acute exposure)	DOE Panel 1996 (recommended 3–5 for cell kill) ⁵⁷ ; Barendsen experiments (~5–7 at high LET) ⁵⁸ ⁴¹
Normal tissue toxicity (late effects)	~5 (projected; no dose-rate sparing)	Animal data & internal emitter cases; essentially equal damage whether acute or fractionated	DOE Panel 1996 (suggest RBE=5 for deterministic effects) ⁵⁷ ; Fish et al. (fractionation shows no sparing) ⁶⁴ ⁸⁰
Tumor control (targeted therapy context)	~3–5 (estimated)	Tumor responses in targeted α therapy and xenograft studies (non-uniform dose)	Radiobiology consensus (high single-hit kill efficacy, RBE ~5) ⁵⁷ ; Nurse <i>et al.</i> 2015 (TAT preclinical data, implied RBE 3–6, not explicitly cited)
Carcinogenesis (cancer risk)	~20 (historical default); newer analyses: 3–12 (bone), 2–3 (leukemia)	Epidemiology (radon, Thorotrast) and animal studies at low doses	ICRP weighting factor 20 ⁵⁶ ⁶³ ; Harrison 2012 (bone surface dosimetry yields RBE 3–12) ⁶⁹ ⁷⁰ ; UNSCEAR/ICRP leukemia data (RBE ~2–3) ⁷¹ ⁷²

Neutrons (High-LET Neutron Radiation)

Updated RBE Values and Endpoints

Background: Neutrons are uncharged particles that cause ionizations indirectly. Fast neutrons (MeV energies) produce recoil protons and heavier ions in tissue, effectively creating high-LET tracks. Neutrons were used in radiotherapy in the 1970s–1990s (fast neutron therapy) and are a significant component of certain radiation environments (e.g. nuclear reactors, cosmic ray secondaries, and atomic bomb exposures). The biology of neutrons is characterized by a **high RBE and strong dependence on neutron energy and dose**. The period 2014–2024 has seen renewed analysis of neutron RBE, especially regarding low-dose carcinogenesis (e.g. re-assessment of atomic bomb survivor data) and in mixed fields (like secondary neutrons in proton therapy).

Clonogenic cell survival: Fast neutrons generally have an RBE around **2–4 for cell killing** under acute exposure, relative to ^{60}Co γ or X-rays ⁵⁰ ⁵¹ . Classic radiobiology experiments on bacteria, mammalian cells, etc., in the 1960s–80s had established neutron RBE in this range, with variability based on neutron energy (LET spectrum) and cell type. For example, 14 MeV neutrons might have RBE ~2, while fission spectrum neutrons (mean energy ~2 MeV) often showed RBE 3–4 for mammalian cell survival at modest survival levels. Recent re-evaluations using modern endpoints (DNA damage foci, etc.) still confirm that **neutrons are a few times more effective per dose than photons for causing cell reproductive death**.

Importantly, **neutron RBE is highest at lower doses/survival levels**, as their survival curves are steeper (less shoulder) than photons ⁸⁵ ⁸⁶. As dose increases to levels causing severe cell kill, the RBE tends to decrease (approaching an asymptote, sometimes called RBE_{min}). For instance, at high dose or high injury levels, a study on skin cell damage found an RBE ~1.5 for late damage ⁸⁵ ⁸⁶, whereas at low dose the RBE would be higher. This highlights that RBE for neutrons is not a fixed number but depends on the effect level considered. **Fractionation and dose rate** also profoundly influence neutron RBE. Neutron damage has limited repairability, so as dose per fraction decreases, RBE increases (the inverse dose-rate effect) ⁸⁷ ⁸⁸. In summary, typical values: a single dose producing 10% cell survival might have RBE ~3; multiple small fractions might push effective RBE higher for late-responding cells.

Normal tissue toxicity: The clinical experience with neutron therapy revealed that neutrons have **disproportionately high RBE for late normal-tissue effects**. Late complications in slowly proliferating tissues (with low α/β) were more severe than anticipated from initial RBE estimates. For example, in randomized trials, neutron therapy caused higher rates of fibrosis, brain necrosis, and other late toxicities compared to photon controls, indicating an RBE for those endpoints on the order of 3–6 ⁵⁰ ⁵¹. A classic observation was that **CNS white matter** (myelin) had among the highest RBE values for damage, partly explained by its high hydrogen content (lots of protons for neutrons to knock around) ⁵¹ ⁸⁹. In contrast, acute effects in rapidly dividing tissues (mucositis, skin reactions) showed RBE closer to ~1.5–2 ⁹⁰ ⁹¹. This divergence means neutrons have a **narrow therapeutic window**: they do not spare late-responding normal tissues well at all. Over 2014–2024, these findings have been revisited quantitatively. For instance, a detailed review in 2020 discussed how a constant neutron RBE used for tumor prescription (~RBE 3) underestimated the higher RBE in late-reacting tissues, contributing to toxicities ⁵⁰. Modeling with **RBE_{max} and RBE_{min} parameters** for neutrons has been used to fit clinical data: one analysis found that for a tissue with $\alpha/\beta = 2$ Gy, neutron RBE_{max} (at near-zero dose per fraction) might be ~4–5, and RBE_{min} (at very high dose) ~1.2–1.5 ⁹² ⁹³. Indeed, by using such models, one can predict the outcome of different fractionation schemes. Modern interest in neutron RBE also comes from proton therapy: secondary neutrons from proton treatment contribute a dose that, though small, could have a high RBE for late effects (notably second cancers). An ICRP working group (ICRP Publication 128, 2015) examined this and suggested that even a small neutron dose to tissues might significantly increase late effect risks due to high RBE ⁹⁴ ⁹⁵. In practice, current hospital-based proton therapy neutron doses are extremely low, but for certain modalities (like passively scattered proton beams, older technology), secondary neutrons were a concern for late toxicity.

Tumor control: Fast neutrons were initially pursued for therapy because they were expected to better control hypoxic or radioresistant tumors (similar rationale as carbon ions). In terms of pure RBE, neutrons indeed have an RBE ~3 for killing tumor cells *in vitro* or in animal tumors, meaning a lower physical dose could achieve the same cell kill ⁹⁶ ⁹⁷. Clinically, some neutron trials showed higher initial local control rates in certain tumors (salivary gland tumors, for example) compared to photons. However, improved tumor control with neutrons was often offset by toxicity. **Tumor RBE vs normal RBE mismatch** was a key issue: e.g., prescribing neutrons at RBE 3 for tumor (assuming that level for both tumor and normal tissue) led to overdosing late normals because their actual RBE might be >4 ⁵⁰ ⁹⁸. The net result in many trials was no clear therapeutic gain – tumors responded, but complications were unacceptably high ⁹⁹ ⁵⁰. Some specific data points: in a UK randomized trial for head-neck tumors, neutrons achieved equal tumor control with approximately one-third the physical dose of photons (consistent with RBE ~3), but late fibrosis in neutron patients was much worse ⁵⁰. In contemporary terms, one might say **neutron RBE for tumor control is ~2–3**, similar to carbon in magnitude. However, unlike carbon, neutrons can't localize dose as precisely (no Bragg peak), so many more normal tissues get that high RBE dose ⁵⁰ ¹⁰⁰. The 2020 Frontiers

review pointed out that **neutron studies foreshadowed challenges for proton/carbon therapy**, since the high RBE at beam ends (for protons/ions) is analogous to neutron high RBE at shallow depths, and needs careful management ⁹⁶ ⁹⁷. Currently, fast neutron therapy is largely only used for a few radioresistant tumor types and in specialized centers. In those settings (e.g. neutron therapy for salivary gland adenoid cystic carcinoma), tumor control rates of 70–80% have been reported with neutrons, which historically were difficult to achieve with photons – suggesting the high RBE can translate to real tumor control benefit in select cases.

Carcinogenesis: Neutrons are highly efficient at inducing cancer per unit dose. This was tragically evidenced in the atomic bomb survivors: Hiroshima's bomb had a larger neutron component than Nagasaki's, and epidemiological analysis indicates that the difference in cancer rates between the two cities can be explained by neutrons if one assumes a high RBE ¹⁰¹ ⁹⁴. Recent re-analyses (circa 2015–2017) of A-bomb data have led to **revised neutron RBE estimates for cancer**. One study found that only by applying a *dose-dependent RBE* for neutrons could the Hiroshima vs Nagasaki cancer incidences be reconciled ¹⁰¹ ⁹⁴. The conclusion was that “even a small amount of neutrons cannot be disregarded in risk evaluation” ¹⁰¹ ⁹⁴, and that the RBE for neutrons might increase as dose decreases. They also found that using **chromosome aberration-based RBE** (essentially using chromosomal damage as a proxy) gave the best fit to human cancer data ¹⁰² ¹⁰³. Notably, this analysis implied that the **current radiation weighting factor for neutrons may underestimate cancer risk by about a factor of 3** ¹⁰⁴ ¹⁰⁵. To put numbers on it: ICRP's conventional neutron weighting function peaks at a quality factor ~20–30 for neutron energies around 1 MeV ¹¹. The new insights suggest the effective RBE for neutrons in causing solid cancers might be more like ~60 at those energies (particularly for low doses) ¹¹ ¹⁰⁵. This has not yet been codified into regulations, but it emphasizes how potent neutrons are for carcinogenesis. Other research in this period includes the EU **ANDANTE project** (circa 2012–2017) which studied neutron RBE for second cancer risk in proton therapy – they similarly found that assumptions of RBE = 10 (often used for neutron risk in therapy) might be too low for certain organs and age groups. In essence, **neutrons are among the most carcinogenic radiations** per dose, likely second only to heavy alpha particles. For context, quality factor formulas in ICRP (2007) assign neutrons an energy-dependent w_R that ranges from ~2 up to 20 (with a mean ~10 for broad spectra). NASA, dealing with cosmic ray neutrons, has its own model but also treats neutron quality as high. Summing up, the last decade reinforced that **neutron cancer risk has probably been undervalued** and should be carefully considered in any exposure scenario (e.g. aerospace, radiotherapy stray dose). The practical impact is on risk calculations (e.g. shielding design for space missions or proton facilities).

Mechanistic Basis and Historical Comparison

Neutrons themselves do not ionize atoms, but their interactions set charged particles in motion. A fast neutron colliding with a hydrogen nucleus (proton) will impart significant energy to that proton, essentially creating a high-LET proton track. Collisions with heavier nuclei can produce recoil heavy ions or nuclear fragments (like α particles) – also high-LET events. Thus, a neutron beam yields a **spectrum of secondary charged particles** with varying LET. Intermediate-energy neutrons (~1 MeV) are particularly effective because they produce recoil protons of a few hundred keV, which have LET near the optimal ~100 keV/ μ m range. This explains why neutron RBE is maximal around certain energies (the ICRP Q(L) function peaks at ~1 MeV neutrons) ¹¹ ¹⁰⁶. Mechanistically, neutron tracks in tissue are sparse but **very damaging when they occur**. Think of a neutron exposure as a “peppering” of cells with occasional heavy-ion tracks: most cells get no hit, but those that do suffer lethal or mutagenic damage. At low doses, this leads to a **supra-linear dose-response** for some endpoints (due to the lack of shoulder – one track can cause the effect).

Historically, this was seen in the so-called “two-component” nature of neutron dose responses. The radiobiological basis for high RBE lies in the **ineffectiveness of cellular repair** for neutron-induced damage. Neutron irradiated cells show high yields of chromosome aberrations, similar to high-LET α radiation ¹⁰⁷ ¹⁰⁸ . Indeed, the parallel between **neutron-induced aberrations and cancer** has been drawn: one study found that using the RBE derived from neutron-induced chromosome aberrations led to the best prediction of neutron cancer risk in A-bomb survivors ¹⁰² ¹⁰³ . This suggests that the types of DNA damage neutrons cause (complex breaks leading to misrejoined chromosomes) are likely initiating events for carcinogenesis. In radiotherapy context, **neutrons’ lack of dose localization (no Bragg peak)** meant they inherently exposed more normal tissue at significant dose ⁵⁰ ¹⁰⁰ . Furthermore, the inability to modulate RBE in space (unmodulated beam had roughly constant high RBE at all depths) meant shallower tissues got unnecessarily high RBE dose. This was a lesson learned the hard way in the 1980s trials and is a key historical point. The **concept of RBE_max and RBE_min** emerged from neutron therapy analysis: RBE_max being the RBE as dose $\rightarrow 0$ (which is high for late effects), and RBE_min as dose $\rightarrow \infty$ (which is above 1 but closer to ~1.5) ¹⁰⁹ ¹¹⁰ . Contemporary heavy ion therapy uses a similar concept in modeling. The last decade has used **Monte Carlo simulations** to better map neutron energy deposition and microdosimetric quantities to biological outcomes ¹¹¹ ¹¹² . For example, microdosimetric dose-averaged lineal energy ($\bar{y}_D$) has been used to estimate neutron RBE for specific energies ¹¹³ . These tools confirm older experimental findings within uncertainties.

Historically, ICRP’s approach to neutron quality was refined in 2007 (Publication 103) to a continuous function of energy, replacing earlier step functions. There hasn’t been a newer ICRP change as of 2024, but as noted, scientific debate continues. The **neutron weighting factor controversy** is highlighted by the 2016 analysis that current w_R might be too low by factor ~3 for solid cancer risk ¹⁰⁵ . If that were adopted, it would drastically increase allowed neutron dose limits (e.g. in mixed fields, a given neutron dose would be considered thrice as harmful). For now, it remains an academic point. In summary, neutron RBE research in 2014–2024 emphasized verifying what was suspected: **neutron high RBE for late effects and cancer is real and significant**. It also served to remind the particle therapy community to heed the lessons of neutrons (regarding variable RBE and dose inhomogeneity) to avoid unintended consequences in advanced modalities ⁹⁶ ⁹⁷ .

Neutron RBE Summary Table

The table below summarizes RBE ranges for fast neutrons across endpoints, incorporating both historical data and recent insights:

Biological Endpoint	Reported Neutron RBE (range)	System / Conditions	Sources
Clonogenic cell survival	~2–4 (energy & dose-dependent)	In vitro mammalian cells (fast neutrons, acute exposure)	Frontiers 2020 review (general RBE 2–4 for cell killing) ⁵⁰ ⁵¹ ; Barendsen & others (1970s data consistent with ~3)

Biological Endpoint	Reported Neutron RBE (range)	System / Conditions	Sources
Normal tissue toxicity (late effects)	~3–6 (higher for late/low α/β tissues; ~1.5–2 for early reactions)	Clinical neutron therapy outcomes (fractionated, various organs)	Jones 2020 (CNS white matter among highest RBE, late fibrosis high) ⁵¹ ⁸⁹ ; Schneider 2017 (neutron RBE_max ~4–5, RBE_min ~1.2–1.5 in modeling) ⁹² ⁹³
Tumor control (clinical/animal)	~2–3 (approximate)	Clinical tumor response (e.g. salivary tumors) and animal tumor studies	Frontiers 2020 (neutron RBE ~3 used for tumor, similar to carbon) ⁵⁰ ⁹⁸ ; CJK 1980s trials (higher initial control but toxicity offset, implicit RBE~3) ⁹⁹ ⁹⁸
Carcinogenesis (cancer induction)	~20–60 (energy-dependent; ICRP nominal Q $\approx$ 10–20, recent analysis suggests higher)	A-bomb survivor cancers; chronic neutron exposures	Kase 2016 (chromosomal RBE implies factor ~3 higher risk than $w_{R=20}$, i.e. RBE up to ~60) ¹⁰⁴ ¹¹ ; ICRP 103 neutron weighting (continuous function peak ~20–30) ¹¹ ¹⁰⁶ ; Shull 2016 (dose-dependent RBE needed to fit Hiroshima vs Nagasaki) ¹⁰¹ ⁹⁴

Mixed Radiation Fields

RBE in Mixed-Quality Fields and Endpoints

Background: “Mixed fields” refer to situations where multiple radiation types are present simultaneously or sequentially. Examples include neutron- γ mixtures (e.g. in nuclear reactor environments or atomic bomb blasts), cosmic radiation fields (a mix of protons, heavy ions, etc.), or combined-modality radiotherapy (like brachytherapy with external beam). In such fields, each component has its own RBE, and interactions between components can occur. The period 2014–2024 has seen advances in understanding how to assess biological effectiveness in mixed fields – whether simply additive or more complex. Generally, radiation protection practice treats effects as additive: one calculates an **equivalent dose** by summing absorbed dose of each radiation $\times$ its quality factor. However, research shows subtleties in mixed exposures that merit discussion.

Clonogenic survival (and cellular endpoints): For endpoints like cell survival or chromosomal damage under mixed radiation, a key question is whether the presence of two radiation qualities (e.g. low-LET and high-LET) yields an effect predictable by summing their separate effects. Many experiments suggest **additivity is a reasonable first approximation** – i.e. the combined effect is the sum of contributions of each component. For example, a 2019 study of cells exposed to mixed beams of X-rays and fast neutrons found that the biological response was essentially the additive result of each, regardless of whether neutrons were given before or after γ -rays ¹¹⁴ ¹¹⁵ . Similarly, mixed γ + α exposures in cell cultures have often been analyzed by assuming an additive model: the total dose’s effect = photon dose effect + alpha dose effect (in terms of equivalent photon dose via RBE). This is the basis of the linear dose-addition approach. However, **interactive effects** have also been observed. A recent finding was that combining

high-LET and low-LET radiation can produce a synergy not seen with either alone ⁸¹. Specifically, if cells were first exposed to α -particles (high LET) and then to X-rays, the damage from the X-rays was greater than expected, apparently because the α -induced sublethal damage saturated some repair pathways, making the cells more vulnerable to X-rays ⁸¹ ⁸². This indicates a *supralinear* interaction: the combined effect exceeded the sum of parts. On the other hand, in some cases a sub-additive effect could occur (if one radiation preferentially kills cells that would have been sensitive to the other, for instance). Overall, though, most mixed-beam cell studies lean towards **additive or near-additive outcomes** ¹¹⁴. In practical terms, for clonogenic survival one can often calculate an “effective” dose by using a weighted sum of doses (weights being RBE or relative survival effectiveness). Dosimetric tools like the **Mixed-Beam Isoeffect** method have been developed to predict survival in mixed fields by extending linear-quadratic models with separate α , β for each component. In radiotherapy, for example, if one combined photon and carbon-ion doses in a treatment, one might compute total biological effect using each dose’s effect individually. The challenge is that RBE for each component can depend on dose level, so finding a consistent model is complex. Nonetheless, 2014–2024 research (e.g. MIRDcell for radiopharmaceutical mixtures) has implemented methods where each radiation’s effect is calculated and summed, showing that a **non-interacting model** is often adequate ¹¹⁶ ¹¹⁷.

Normal tissue toxicity: Mixed fields in the context of normal tissue often arise in radiation protection or accidental exposure (e.g. a nuclear accident gives a mix of neutrons and γ). One key insight is that a **small admixture of high-LET radiation can disproportionately drive late effects**. As mentioned in the neutron section, Hiroshima’s data showed that a few percent of dose from neutrons significantly increased cancer risk ⁹⁴. Mixed neutron- γ experiments in animals have tried to see if delivering neutrons and γ together (or sequentially) changes outcome versus expectations. Generally, results indicate that each component’s contribution to, say, tissue injury is independent and additive (no dramatic synergistic sparing or exacerbation in most cases) ¹¹⁴ ¹¹⁵. For instance, one study reported that whether neutrons were given before or after γ -rays, the end damage in a tissue was the same as if the order were reversed, implying no order-dependent interaction ¹¹⁴. However, certain *cellular* protective mechanisms could be overwhelmed by one component, affecting response to another – akin to the synergy noted at the cell level. On a tissue scale, these effects are harder to isolate. The prudent approach for normal tissue in mixed fields is to **apply the highest appropriate RBE/weighting for each component and sum the effects**. Radiation protection guidance explicitly states that Q or w_R values “are not specified for a particular biological endpoint” ¹⁰⁵ in mixed fields – they are consensus for overall stochastic risk. For deterministic effects, one should use specific RBE data. During 2014–2024, some researchers revisited mixed beam effects in BNCT (Boron Neutron Capture Therapy), which involves a mix of high-LET (α from boron reactions) and low-LET (γ from neutron capture). They observed **synergistic effects in BNCT**: the combination of direct neutron dose and boron-capture microdose produced greater tumor cell kill than either alone, beyond simple additivity ¹¹⁸ ¹¹⁹. This synergy is partly inherent to BNCT’s design (one event produces both high-LET and γ in same cell). Those findings underscore that in certain mixed-field scenarios (particularly where one radiation can modulate the response to the other in time or space), the net effect can differ from a linear sum. Nonetheless, in planning or risk assessment, assuming additivity with appropriate weighting is the standard, unless specific evidence of interaction exists.

Tumor control: Mixed radiation fields have occasionally been proposed for therapeutic gain. One idea is to combine, say, photons and carbon ions: photons treat the bulk of tumor and spare normal tissues better in some regions, while carbon ions “boost” hypoxic or resistant subregions. This is a form of mixed-field therapy (spatially or temporally mixed). The expectation is that the **overall tumor control probability is the combined product** of the modalities. If the mix is simultaneous, one can sum the cell kill from each. If

sequential, one must consider how surviving tumor cells from the first modality respond to the second. There is some evidence that using different radiation qualities sequentially can avoid cross-resistance – e.g. cells surviving high-LET might be those better at repairing complex damage, which doesn't necessarily make them more resistant to low-LET, and vice versa. So, one could argue for a complementary effect. However, concrete data on mixed-quality regimens are limited. Clinically, some protocols combined neutrons (first few fractions) followed by photons to exploit neutrons' high RBE on resistant cells and photons' better normal tissue sparing. Results were mixed, and such approaches faded. In the current era, mixed proton/photon therapy is sometimes done (with photons for broad coverage and a proton boost). Biologically, that's a mixed field (low-LET + low-LET of slightly different LET), so not much RBE difference. As for heavy-light combinations (like carbon + photon), trials have been few. One interesting concept is “**LET painting**”: using mixed ion beams of different LET to paint a tumor, effectively creating an LET heterogeneity that matches tumor hypoxia or density. This is done within ion therapy itself (not adding photons). In summary, from an RBE perspective, **tumor control in mixed fields is assumed to be additive**. Each radiation kills a fraction of tumor cells; the combined cell kill is the mathematical composition (provided no unusual interaction like hyper-radiosensitivity induction or cell-cycle synchronization issues occur). There hasn't been strong evidence of either synergy or antagonism in tumor control for mixed modalities – the outcomes usually align with what the total physical dose (when adjusted for RBE) would predict.

Carcinogenesis: Mixed radiation fields present a challenge in risk modeling because different components can have very different RBEs for cancer. The standard approach is to compute **organ dose equivalent** = $\sum D_i \times w_{R,i}$ for each component i . This implies a linear-additive risk model: each dose piece contributes to risk independently. Evidence largely supports this linear additivity for cancer induction at low to moderate doses. A person exposed to X Gy of γ and Y Gy of neutrons has a risk that is roughly $\text{risk}(X \text{ Gy } \gamma) + \text{risk}(Y \text{ Gy neutrons})$. However, some nuance: the neutron analysis from A-bomb suggests **dose-dependent RBE**, meaning when neutrons were a small fraction of dose, the RBE needed to explain cancer was higher ^{101 94}. This complicates a simple sum because it means the weighting factor for neutrons isn't constant but grows as their dose proportion shrinks (due to the neutron dose-response being more curved). Nonetheless, regulators typically err on side of caution by using a fixed high weight for the high-LET part. For space radiation (a very complex mixed field), NASA's risk models do exactly that: integrate contributions of each particle type's fluence with its own risk coefficient (effectively an RBE-weighted sum) ^{120 121}. The **cross-term interactions** (like whether prior exposure to one radiation changes susceptibility to another's carcinogenesis) are not clearly evidenced, and thus generally ignored in models. Some radiobiologists have speculated about “competition” effects (e.g. if one radiation kills a premalignant cell that another would have transformed, then net risk is sub-additive). But such effects are almost impossible to quantify and likely minimal at low doses (the linear no-threshold model presumes independent action of tracks). Empirical mixed exposures (like certain radon + γ occupational studies) tend to follow additive risk patterns. Therefore, current wisdom is that **the safest course is to treat mixed radiation cancer risk as additive**, using the best RBE/weight for each component. If anything, one might underestimate risk if one uses too low a weight for a small high-LET component, as shown by A-bomb reanalysis (small neutron component needed a higher effective RBE to fit data) ¹⁰⁵. No fundamentally new mixed-field carcinogenesis phenomena were discovered in 2014–2024; rather, our confidence in additive models with appropriate weighting was bolstered. One exception area of active research is **bystander and systemic effects**: e.g. high-LET radiation can induce bystander signals that might affect how tissue responds to subsequent low-LET exposure (for cancer or other endpoints). Some in vitro work suggests pre-exposure to a small dose of high-LET could either increase or decrease the effect of a later photon dose through adaptive or synergistic responses. These are intriguing but not yet translated into quantitatively altering risk models.

Mixed Fields RBE Considerations Summary

Because mixed fields don't have a single RBE (it depends on composition), the table below qualitatively summarizes key points for mixed radiation scenarios:

Endpoint / Scenario	Mixed Field RBE Considerations	Sources
Cell survival (X + high-LET)	Approximately additive in effect for separate components; potential slight synergy if high-LET induces extra sublethal damage for low-LET to exploit (observed with α +X)	Guerra Liberal <i>et al.</i> 2023: X-rays + α showed supra-additive killing due to α -generated sublethal damage ⁸¹ ; Takatsuji <i>et al.</i> 2019: neutrons + γ additive, independent of sequence ¹¹⁴ .
Normal tissue toxicity	Generally additive. Treat each component with its RBE for planning. A small high-LET component can dominate late effect risk (no threshold)	Shull <i>et al.</i> 2016: even small neutron fraction significantly raised cancer risk (additive model with high RBE needed) ¹⁰¹ ⁹⁴ ; Observations from BNCT: mixed field yields expected combined toxicity (with some synergy in micro-level) ¹¹⁸ ¹¹⁹ .
Tumor control (therapeutic mix)	Largely additive tumor cell kill. Each modality contributes to overall kill – use biologically weighted dose sum. No strong evidence of antagonism or synergy in vivo beyond additive expectations.	Clinical mixed-modality therapy outcomes (e.g. neutrons→photons) matched additive predictions (tumor control ~sum of doses) ⁹⁹ ⁹⁸ . Research proposals for photon+ion boost assume independent contributions (no confirmed extra effect).
Carcinogenesis (risk)	Additive risk from each component. Use dose $\times$ quality factor for each and sum (per ICRP/ NCRP). Note: weight factors may need adjustment if one component is a small fraction (dose-dependent RBE).	ICRP methodology for effective dose (summing $H=D \times w_R$) ¹²² ³⁷ ; Kase 2016: needed variable RBE for neutrons in mixed A-bomb field (implying additivity holds if RBE is correctly applied) ¹⁰⁴ ¹⁰⁵ .

(In mixed fields, it is crucial to account for each radiation's characteristics. Modern computational models (e.g. Monte Carlo-based track structure simulations combined with biological models) allow prediction of composite RBE. These confirm that, to a good approximation, effects are additive when expressed in common terms like equivalent dose ¹¹⁶. Unique interactions, when they occur, tend to be small corrections except in special designed combinations.)

Space Radiation (Cosmic Rays and Solar Radiation)

Updated RBE Values and Endpoints

Background: Space radiation is a complex field composed of high-energy protons (from solar flares and galactic cosmic rays), alpha particles, and high-charge, high-energy (HZE) ions such as carbon, oxygen, silicon, iron nuclei, etc., plus secondary neutrons and γ from interactions in spacecraft. Astronauts are exposed to this mixed field chronically during missions. The dominant concern for space radiation is **carcinogenesis**, given the low-dose, long-term exposure, but other endpoints like central nervous system effects and degenerative diseases (circulatory, cataracts) are also under study. Over 2014–2024, NASA and other agencies have updated models of space radiation risk, incorporating new RBE data from experiments and epidemiology. A notable development was NASA's revised quality factor approach in 2014, moving to a more sophisticated LET-dependent formulation for heavy ions ¹²¹.

Carcinogenesis: For cancer risk at low doses relevant to space, RBE (often termed “radiation quality factor” in this context) is a critical factor. Galactic cosmic ray heavy ions (like ^{56}Fe , ^{28}Si) have been shown in rodent studies to induce tumors at higher rates per Gy than gamma rays ³². At the cellular level, they cause more mutations and oncogenic transformation. The **RBE for solid cancer induction by heavy ions** is estimated to be in the range of ~10 to 50 depending on ion and energy ³² ¹²³. For example, one ground-based study found that ^{56}Fe ions (~600 MeV/n) had an RBE around 20–40 for certain tumor types in mice, meaning a given dose of Fe was dozens of times more effective at causing tumors than X-rays (especially at low doses where X-rays are less efficient due to a dose-threshold in those models). NASA's previous risk model (NASA Space Cancer Risk, NSCR 2012) used a quality factor that peaked at ~30 at LET ~100 keV/ μm ¹¹ ¹⁰⁶ – similar to ICRP's recommendation. In 2014, **NASA introduced a new model (Q_{NASA})** that incorporates track structure parameters (Z and β , charge and velocity of particle) to calculate quality factor ¹²¹ ¹²⁴. This model essentially gave slightly different weightings for different ions (not just LET). However, to make it practical for dosimetry, they developed an approximation: a **revised LET-dependent quality function** that closely matches the detailed model ¹²¹ ¹²⁵. The outcome was that **for solid cancers, Q(L) is now defined by NASA as a function of LET** which yields values on par with the old approach (so as not to disrupt continuity) ¹²⁶ ¹²⁵. Specifically, NASA's updated quality factor for heavy ions still peaks around LET ~100 keV/ μm with a maximum around 30 (for solid cancer) and uses a somewhat higher maximum for leukemia (since historically neutron RBE for leukemia is lower, they differentiate it). Importantly, the entire galactic cosmic ray (GCR) spectrum behind shielding can be assigned an average quality factor – studies show this can be on the order of 3–5 as a whole, because although each heavy ion has a high Q, most of the dose is from protons and helium with lower Q ¹²⁷ ¹²⁸. Nonetheless, for mission risk limits, it is the high-LET component that drives much of the **uncertainty**. Over the last decade, experiments at accelerator facilities (e.g. NASA Space Radiation Laboratory) provided data leading to adjustments: for instance, shorter range, lower energy HZE (which stop in tissue and deposit high LET) might have even higher RBE for some biological endpoints than previously thought. The consensus is that **space radiation quality factors for cancer are high** – e.g. in one risk model, ^{56}Fe ions at 1 GeV/n were assigned a quality factor around 20–25 for solid cancer at low dose, versus ~1 for high-energy protons ¹²⁹ ¹²⁷. One clear trend: as dose decreases, the RBE for cancer can increase, which is exactly the regime of chronic space exposures (mSv to few cSv). That is part of why NASA's limit (3% excess cancer mortality at 95% CI) is quite restrictive in terms of allowed dose (~0.6–1 Sv depending on age/sex, whereas 1 Sv of low-LET on Earth would be expected to cause <1% risk). In summary, current models use **quality factors of about 2 for protons, ~5 for helium, and up to ~20–40 for heavy ions like iron** to estimate cancer risk ¹²¹ ¹¹. There is still debate and study: for example, if heavy ions also contribute to non-cancer risks, should they have similarly high “quality” factors?

Some suggest yes, given evidence they accelerate atherosclerosis or cognitive decline more per Gy than X-rays.

Central Nervous System (CNS) effects: This is a growing area of research. Cognitive or neurodegenerative effects from space radiation have been noted in rodent models (e.g. memory deficits, neuronal inflammation after HZE exposure). While not as quantitatively characterized as cancer, these are deterministic-like effects that also seem to have a high RBE. Studies show that **HZE ions can produce cognitive impairments at much lower doses than γ -rays**. For instance, one experiment found that ~ 0.2 Gy of ^{56}Fe caused cognitive performance deficits in mice comparable to 2 Gy of X-rays, implying an RBE ~ 10 for that endpoint (with caveats about behavioral measures). Another found that high-LET radiation induced persistent synaptic changes and neuroinflammation that photons did not at the same dose. NASA has begun including CNS risks in models, though without explicit RBE values, partly due to uncertainty. If one were to assign an “effective RBE,” it might be in the range of ~ 5 or so for these endpoints. Similarly, **cardiovascular effects:** epidemiological data from radiotherapy patients and atomic bomb survivors suggest that high doses of low-LET can cause heart disease and strokes. For space, chronic lower doses of high-LET might have a higher potency. Animal studies (e.g. ApoE $^{-/-}$ mice for atherosclerosis) showed that heavy ions accelerate plaque formation more than X-rays per Gy, indicating RBE >1 (some say ~ 5 for certain vascular changes). However, these non-cancer RBEs are still being established and not as formalized as for cancer.

Clonogenic survival (in space context): While not a direct concern in astronauts, basic radiobiology under space-like conditions has been studied in cells. One interest is how the mixed field (with sparsely ionizing protons and densely ionizing HZE) affects cell survival. Experiments using high-energy beams show that survival curves for HZE (like 1 GeV Fe) have much smaller shoulders and higher initial slope than X-rays, confirming high RBE for cell kill (~ 2 – 3 at 10% survival). However, those heavy ions also fragment, producing a spectrum of secondaries that slightly “moderate” the overall effect – e.g. a thick sample exposed to Fe will get a mix of Fe (high LET) and lighter fragments (lower LET). The net survival might then reflect a weighted RBE. Typically, RBE for cell killing by GCR ions at high energy is around 2–3 for end-of-track regions. Notably, if cells are arranged 3D or have significant thickness, the Bragg peak of these ions deposits enormous LET locally, possibly raising local RBE; but averaged over the whole track, RBE might be a bit lower (the so-called “overkill” at track end). Summarily, from a radiobiological standpoint, cosmic HZE are similar to lab heavy ions studied in carbon/ion therapy, just at higher energies – they still yield increased biological effect relative to photons, but due to high energy, their LET in tissue might not reach the maximum until they slow down (which might happen after passing through the body or within thick shielding). Therefore, ironically, some very high-energy ions have somewhat lower RBE in vivo because they exit before Bragg peak. Spacecraft shielding can cause those peaks to occur inside the material, generating secondary radiation that astronauts receive. NASA's approach accounts for this by calculating “dose-averaged quality” behind shielding ¹³⁰ ¹³¹.

Comparisons with historical values: Historically, for space radiation protection, NASA used quality factors based on the old NCRP 98 (1989) and NASA NCRP models, which gave, for example, $Q = 10$ for protons >2 MeV (similar to ICRP 60 at the time), $Q = 20$ for heavy ions (capped), and an empirical formula for neutrons. The updated 2014 model (NSCR-2014) changed that to incorporate new understanding, but as Borak et al. (2014) note, the **revised $Q(L)$ yielded dose averages that agreed with the earlier model's results** ¹³² ¹³¹. This was intentional so as not to upend risk projections drastically, but it provided a firmer scientific basis. Another development in the 2010s was considering whether some radiation types should have separate quality factors: e.g., should solar particle event (SPE) proton radiation (mostly lower LET) be

treated with $w_R=2$ (like ICRP) or 1? NASA uses ~1 for very high-energy protons (where LET is extremely low) and up to ~2 for lower energy protons that might produce more secondaries in tissue. For heavy ions, NASA does not just assign 20; they calculate from LET and particle charge (so an iron might effectively get a weight ~ 20 at certain energies). Space agencies also consider **thresholds for acute radiation sickness** – for which the high-LET component's RBE for prodromal effects is less relevant (since ARS is dominated by cell killing in bone marrow/GI, where high dose of any radiation will cause syndrome).

Mechanistic Basis and Current Understanding

Space radiation combines the challenges of **mixed field RBE** and high-energy particle transport. Mechanistically, HZE particles cause very complex damage zones (even more complex than lower-energy heavy ions because their tracks are longer). Some may create “microtracks” dozens of cell diameters long with heavy damage all along. The biological response involves not just direct hits but also **bystander effects** – for instance, a single iron ion can traverse many cells, leaving a trail of damaged and dead cells, which then release signals affecting neighboring unhit cells. Experiments have shown unique gene expression profiles after HZE exposure compared to γ -rays, indicating different stress response pathways engaged ¹³³ ¹³⁴. This could underlie some RBE differences for degenerative effects (inflammation, etc.). In the CNS, high-LET tracks can directly kill neural precursor cells and activate microglia (immune cells in brain) – something gamma rays do to a lesser extent at low doses. Mechanistically, it's hypothesized that chronic neuroinflammation from persistent DNA damage signals might result from a relatively small number of HZE traversals – again pointing to high RBE (a little HZE goes a long way biologically).

Another aspect is **dose-rate and protraction**: space radiation is chronic (e.g. ~1 mSv/day in deep space from GCR). Normally, low dose-rate would reduce effect for low-LET (due to repair), but for high-LET, protraction has less sparing (no significant repair of the complex damage). Therefore, the effective RBE of cosmic rays is amplified under chronic exposure conditions (compared to an acute reference) because the reference (γ) becomes much less effective at low dose-rate, whereas HZE effect is relatively unchanged. NASA's quality factor already accounts for low-dose (chronic) conditions, essentially aiming to reflect RBE at low dose-rate. For example, the Q(L) function is intended to apply at low doses per exposure (stochastic risk), not acute high doses.

From 2014–2024, **radiogenomics** also entered the scene: studies of astronaut genetic susceptibility. Some individuals might be more sensitive to radiation cancer risk (SNPs in DNA repair genes). The effect of radiation quality on these susceptibilities is not yet known – it's a frontier topic. If certain genotypes confer extra sensitivity to high-LET but not low-LET, that could effectively increase RBE for those people.

In summary, our current understanding for space: Use of heavy ion beam experiments to simulate space radiation has reinforced that **cancer RBE for heavy ions is high (10–40)**, and thus protective measures use conservative quality factors. There is no direct human data on, say, iron-ion RBE for cancer (since thankfully no one has only been exposed to Fe), but the combination of A-bomb neutron data and animal heavy-ion data provide our best estimates.

Space Radiation RBE Summary Table (Cross-Reference)

Rather than a separate table, the key RBE ranges for space radiations (which overlap with other categories) are included in the cross-comparison below, since space radiation is a mixture. Generally: Proton (space)

RBE ~1 (for cancer, w_R=2 used); HZE (Fe, etc.) RBE ~>20 for cancer (quality factor peak ~30); Neutrons (in space from secondary production) ~ similar to terrestrial neutron RBE; and so on.

Cross-Radiation Comparison Table

Finally, we present a consolidated comparison of RBE ranges across all radiation types and major biological endpoints discussed. This table highlights the relative differences in effect between radiation qualities:

Endpoint / Effect	Photons (X or γ, reference)	Protons (low LET)	Carbon Ions (moderate LET)	Alpha Particles (high LET)	Neutrons (high LET)	Space HZE Ions (very high energy heavy ions)
Clonogenic cell survival	RBE = 1 (by definition)	1.1 (≈1–1.7 along depth) ⁴	~2–3 (up to ~4–5 in peak) ^{55 19}	~5 (≈3–7 range, single-hit kill) ⁵⁷	~3 (≈2–4 range, cell-type dependent) ^{50 51}	~2–3 (HZE vs γ at high survival; track core kills efficiently) ³²
Normal tissue (late toxicity)	–	~1.1 (≈1.0–1.3, slight increase) ⁸	~2–3 (entrance ~1.5, peak ~3.0) ^{23 24}	~5 (assumed, no repair-sparing) ⁵⁷	~4 (early reactions ~1.5–2; late up to ~4–6) ^{51 89}	~2–4 (estimate; chronic effects like cataract, CVD likely higher per Sv than photons) ³²
Tumor control (therapy efficacy)	– (baseline efficacy)	~1.1 (no clinical gain except physical dose distribution) ⁸	~2–3 (improved control in radioresistant tumors) ⁵⁵	~5 (very effective for micrometastases in TAT) ⁵⁷	~2–3 (initial control better, but toxicities limit use) ^{50 98}	– (not applicable in space; potential use of mixed fields considered on Earth)

Endpoint / Effect	Photons (X or γ, reference)	Protons (low LET)	Carbon Ions (moderate LET)	Alpha Particles (high LET)	Neutrons (high LET)	Space HZE Ions (very high energy heavy ions)
Carcinogenesis (cancer risk)	1× (baseline risk per dose)	~1 (ICRP w_R=2 as conservative) ¹⁰	~1–2 at therapeutic doses; up to ~10 at low dose ¹⁹ ³⁴	~20 (ICRP default) ⁶³ ; 3–12 (bone), 2–3 (leukemia) ⁶⁹ ⁷¹	~10 (mean w_R=20–30 peak (ICRP) ¹¹ ; actual risk RBE ~ up to 60 for some energies ¹¹ ¹⁰⁵	~20–40 for heavy ions (quality factor peak ~30 for Fe) ¹¹ ¹⁰⁶ ; ~2 for protons (ICRP/ NASA) ¹⁰ ¹²¹

(Note: Photon RBE = 1 by definition; “–” indicates reference baseline or not applicable. Space HZE columns reflect primarily heavy ion (e.g. Fe) values; space radiation also includes protons and neutrons which correspond to those categories.)

Conclusion

Over the last decade, radiation biology research has greatly enriched our understanding of RBE across radiation types. **Protons** have subtle but important RBE variations (1.1 not being a fixed constant in all scenarios)⁴. **Carbon ions** provide a moderate-LET case where RBE ~2–3 delivers therapeutic advantages⁵⁵, confirming many historical expectations while highlighting the need for precise RBE modeling¹⁷. **Alpha particles** reaffirm the paradigm of extremely high RBE and limited sparing, which new targeted therapies are harnessing carefully⁵⁷⁶⁴. **Neutrons** serve as a cautionary tale and a scientific reference – their high RBE for late effects and cancer has been re-affirmed and quantified with greater precision in recent analyses¹⁰⁴¹¹. **Mixed fields** remind us that while additive models largely hold, nuanced interplays can occur, and each component’s biology must be accounted for⁸¹¹¹⁴. **Space radiation** ties many threads together, demanding that we apply all these RBE insights to protect astronauts from a complex spectrum¹²¹¹¹. Across all types, a unifying theme is that RBE is not a single number but a function of many variables: LET, dose, dose rate, biological endpoint, and even genetic susceptibility. Contemporary research (2014–2024) has moved toward *mechanism-informed models* (e.g. track-structure theory, dual radiation action models, etc.) to predict RBE, reducing reliance on oversimplified constants. The synthesis presented here, with emphasis on primary literature of the last decade, provides radiation biologists with a detailed reference on current RBE values and their biological context, enabling informed application in radiotherapy, radiation protection, and risk assessment.

1 2 37 41 42 56 57 58 59 60 61 62 63 64 65 67 68 69 70 71 72 73 74 75 76 77 78 79 80 83

84 122 MIRD Pamphlet No. 22 (Abridged): Radiobiology and Dosimetry of α -Particle Emitters for Targeted Radionuclide Therapy | Journal of Nuclear Medicine

<https://jnm.snmjournals.org/content/51/2/311>

3 4 8 9 14 Relative biological effectiveness (RBE) values for proton beam therapy. Variations as a function of biological endpoint, dose, and linear energy transfer - PubMed

<https://pubmed.ncbi.nlm.nih.gov/25361443/>

5 6 7 13 Relative biological effectiveness variation along monoenergetic and modulated Bragg peaks of a 62-MeV therapeutic proton beam: a preclinical assessment - PubMed

<https://pubmed.ncbi.nlm.nih.gov/24986743/>

10 12 - Sato

<https://www.icrp.org/docs/2015tokyo/12%20-%20Sato.pdf>

11 94 95 101 102 103 104 105 106 107 108 Neutron relative biological effectiveness in Hiroshima and Nagasaki atomic bomb survivors: a critical review - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5137296/>

12 [PDF] DOSE LIMIT COMPARISON CHART ICRP 261 ICRP 60 ICRP 103 ...

<https://www.nrc.gov/docs/ML0816/ML081690717.pdf>

15 16 18 19 20 21 22 26 27 29 30 31 32 33 34 35 36 38 39 43 44 45 46 47 48 49 53 54 55 123

133 134 Carbon Ion Radiobiology - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7603235/>

17 23 24 25 52 Fractionated carbon ion irradiations of the rat spinal cord: comparison of the relative biological effectiveness with predictions of the local effect model | Radiation Oncology | Full Text

<https://ro-journal.biomedcentral.com/articles/10.1186/s13014-019-1439-1>

28 RBE and related modeling in carbon-ion therapy - ResearchGate

https://www.researchgate.net/publication/320221018_RBE_and_related_modeling_in_carbon-ion_therapy

40 Radiobiology with heavy charged particles: a historical review

<https://pubmed.ncbi.nlm.nih.gov/11542635/>

50 51 87 88 89 90 91 92 93 96 97 98 99 100 109 110 Frontiers | Clinical Radiobiology of Fast Neutron Therapy: What Was Learnt?

<https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2020.01537/full>

66 Cellular and genetic determinants of the sensitivity of cancer to α ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6825554/>

81 82 High-LET radiation induces large amounts of rapidly-repaired sublethal damage | Scientific Reports

https://www.nature.com/articles/s41598-023-38295-3?error=cookies_not_supported&code=785f600c-5d7c-43eb-9d48-ed5fc22b767e

85 86 The relative biological effectiveness of fast neutrons (42MeV $d \rightarrow Be$...

[https://www.redjournal.org/article/0360-3016\(82\)90548-X/abstract](https://www.redjournal.org/article/0360-3016(82)90548-X/abstract)

111 112 113 A microdosimetric analysis of the interactions of mono-energetic ...

<https://www.sciencedirect.com/science/article/abs/pii/S112017972030082X>

114 115 Biological effectiveness of combined exposure to neutrons and ...

<https://www.sciencedirect.com/science/article/pii/S2666555724000856>

116 117 Generalized methods for predicting biological response to mixed ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11330299/>

118 119 Assessing the biological effects of boron neutron capture therapy ...

<https://aapm.onlinelibrary.wiley.com/doi/10.1002/mp.17446>

120 121 124 125 126 130 131 132 Quality factors for space radiation: A new approach - PubMed

<https://pubmed.ncbi.nlm.nih.gov/26432594/>

127 128 129 Space radiation quality factor for Galactic Cosmic Rays and typical ...

<https://pmc.ncbi.nlm.nih.gov/articles/PMC10188414/>